Google Summer of Code Proposal

Linear Attention Vision Transformers for
End-to-End Jet Image Classification and Mass Regression

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Abstract

This project develops linear-scale attention vision transformer models for end-to-end jet image classification and regression at the LHC. Standard vision transformers suffer from quadratic attention complexity $\mathcal{O}(N^2)$ in the number of image patches, making them expensive at fine granularity. We implement and benchmark the **Cross-Covariance Image Transformer (XCiT)** against a standard ViT architecture.

XCiT achieves a test ROC-AUC of 0.9488 with linear attention complexity $\mathcal{O}(Nd^2/h)$. A self-supervised Masked Autoencoder (MAE) pretraining stage on 30 000 unlabeled jet images (out of 60 000 due to computational constraints) is followed by multitask fine-tuning for simultaneous classification, jet mass regression ($r = 0.875$, MAE = 18.305 GeV), and transverse momentum regression. The p_T result ($r = 0.621$) is interpreted as a physically meaningful signal arising from residual energy-scale correlations surviving the per-sample normalization, rather than a modeling failure.

By combining linear attention, self-supervised learning, and multitask objectives, this project contributes a **scalable and extensible** transformer-based framework that can enhance the E2E project's capability to **perform accurate, interpretable, and computationally efficient event reconstruction** in realistic experimental settings.

The resulting models and training pipeline will be released as reusable components within the E2E ecosystem.

1 About Me

I am an undergraduate Computer Science student at IIT Palakkad with strong interests in scientific machine learning and computational high-energy physics. My background includes coursework in linear algebra, multivariable calculus, probability and statistics, and classical mechanics. I have hands-on experience building deep learning systems in Python using PyTorch, and I actively work on projects that combine deep learning with physical modeling.

For the E2E evaluation task, I implemented XCiT from scratch, designed a complete two-stage training pipeline (MAE self-supervised pretraining on unlabeled data followed by multitask fine-tuning on labeled data), and produced and physically interpreted all results including a non-trivial p_T finding. I am also involved in research at IIT Palakkad on distributed ensemble learning for decentralized datasets, studying theoretical model risk bounds, and building large-scale experimental pipelines.

- **Code Repository:** github.com/Abhiraj212024/E2E-Linear$_Attn_ViT$
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2 Problem Overview

Jet classification is a central challenge at LHC. Jets are collimated sprays of particles produced during hard collisions, and distinguishing jet types is essential for new physics searches. Jet images represent calorimeter energy deposits on an (η, ϕ) grid and encode jet substructure as spatial patterns across 8 channels corresponding to different particle species.

Although convolutional neural networks have demonstrated strong performance, vision transformers offer a more flexible global receptive field. However, standard self-attention scales as $\mathcal{O}(N^2)$ in the number of patches N, making it expensive at fine granularity. This project addresses the efficiency bottleneck by replacing quadratic attention with linear-complexity alternatives, specifically XCiT [1] and as a further direction L^2ViT [2].

3 Proposed Architecture

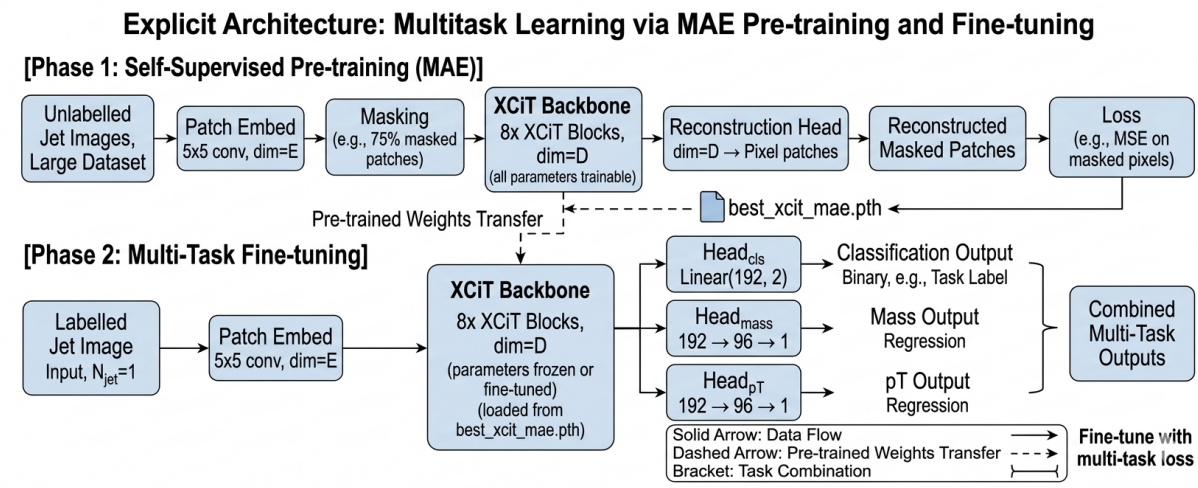

Figure 1: Overall pipeline: MAE pretraining followed by multitask fine-tuning using the XCiT backbone.

3.1 Baselines

- **Standard ViT**: Quadratic self-attention, patch size 5, dim 192, depth 8. Serves as a reference point for the transformer.

3.2 XCiT: Cross-Covariance Image Transformer

XCiT [1] replaces the $N \times N$ token attention map with a $d_k \times d_q$ cross-covariance attention computed over the feature dimension:

$$\mathcal{A}_{\mathrm{XC}}(K, Q) = \mathrm{Softmax}\Big(\hat{K}^\top \hat{Q}/\tau\Big) \in \mathbb{R}^{d_k \times d_q} \tag{1}$$

where $\hat{K}, \hat{Q}$ are ℓ_2-normalized across the token dimension and τ is a learnable temperature. Complexity reduces from $\mathcal{O}(N^2 d)$ to $\mathcal{O}(N d^2/h)$, linear in N. A Local Patch Interaction (LPI) module of two depth-wise 3×3 convolutions provides explicit spatial communication, and class attention layers (CaiT-style) aggregate patch tokens for the classification head.

Configuration. Patch size 5, image 125×125, giving 625 tokens. Embedding dim = 192, depth = 8 XCiT blocks, heads = 8, CLS-attention depth = 2. Total encoder parameters: $\approx$7M.

3.3 Multitask Extension

The backbone is extended with three parallel heads:

$$\hat{y} = \text{head}_{\text{cls}}(\mathbf{z}) \qquad \text{CrossEntropy} \tag{2}$$

$$\hat{m} = \text{head}_{\text{mass}}(\mathbf{z}) \qquad \text{MSE, jet mass} \tag{3}$$

$$\hat{p}_T = \text{head}_{p_T}(\mathbf{z}) \qquad \text{MSE, transverse momentum} \tag{4}$$

Each regression head is a two-layer FFN: $C \to C/2 \to 1$. The combined objective is:

$$\mathcal{L}_{\text{total}} = w_{\text{cls}}\, \mathcal{L}_{\text{CE}} + w_{\text{mass}}\, \mathcal{L}^{m}_{\text{MSE}} + w_{p_T}\, \mathcal{L}^{p_T}_{\text{MSE}} \tag{5}$$

4 Physics Motivation

4.1 Jet Mass Regression

The jet mass depends on pairwise angular separations between the constituents:

$$m^2_{\text{jet}} = \sum_{i,j} p_{T,i}\, p_{T,j} (\cosh \Delta\eta_{ij} - \cos \Delta\phi_{ij}) \tag{6}$$

- Encodes the correlations between the constituents in the (η, ϕ) space.
- Directly reflected in the spatial structure of jet images.
- Transformer backbone learns these spatial patterns during classification.
- These representations transfer naturally to mass prediction.

4.2 The p_T Result and Its Physical Interpretation

Jet images are preprocessed as:

$$\tilde{I}^{(c)}_{ij} = \frac{\log(1 + I^{(c)}_{ij})}{\max_{i',j',c'} \log(1 + I^{(c')}_{i'j'}) + \varepsilon} \tag{7}$$

- Per-sample normalization maps each image to $[0, 1]$.
- Suppresses the absolute energy scale; however, the residual signal remains via the image **dynamic range**.
- The model learns to exploit this residual for the prediction of p_T.
- Observed result: $r_{p_T} = 0.630$.

Conclusion: The non-zero p_T correlation reflects a physically meaningful residual energy-scale signal, not a modeling artifact.

5 Experimental Results

5.1 Phase 1: MAE Pretraining

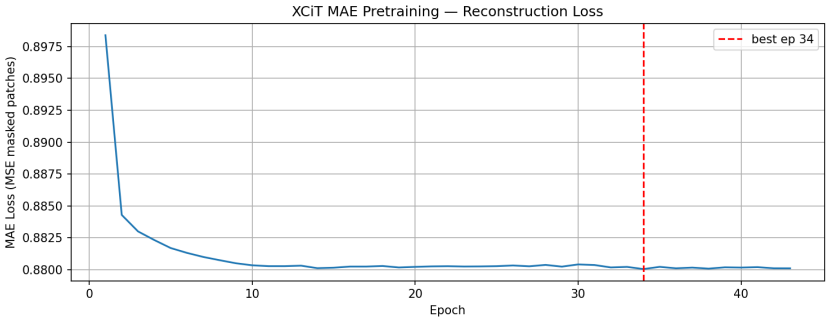

Figure 2: MAE reconstruction loss over training. The early plateau reflects bounded patch-normalized loss behavior on sparse jet images.

- Trained for **50 epochs** on **30k unlabeled jet images**.
- The loss decreases from **0.898** $\rightarrow$ **0.880** and plateaus with **epoch** $\sim$**10**.
- The loss is **bounded in** $[0, 1]$ due to patch-normalized targets.
- Smooth plateau indicates **stable training** on sparse data.
- The model learns **nontrivial spatial structure** beyond zero baseline.

5.2 Phase 2: Multitask Fine-tuning

- Trained on **80% (8k)** samples, evaluated on **20% (2k)**.
- Multitask setup: **classification + mass +** p_T **regression**.
- Results summarized in Table 1.

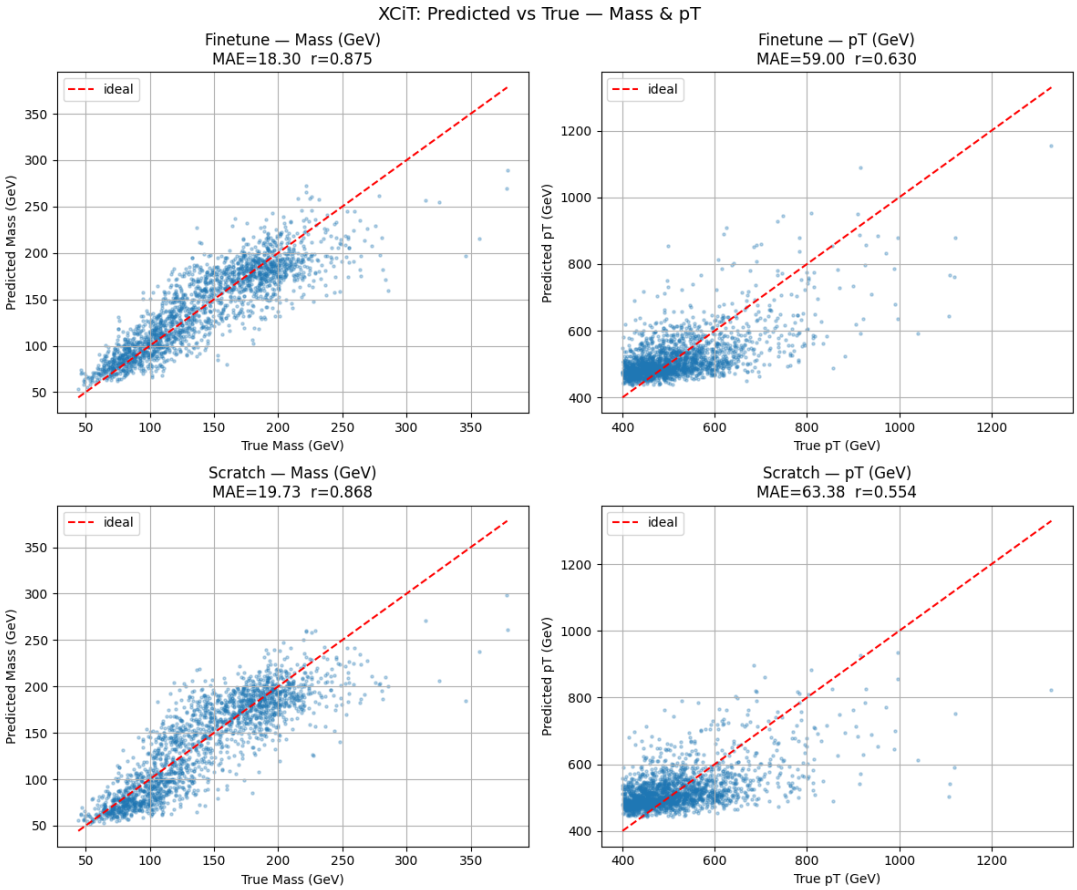

Figure 3: Predicted vs true values for jet mass and p_T. The MAE-pretrained model produces tighter, better-calibrated predictions compared to training from scratch.

Table 1: Multitask performance comparison.

Model	Acc.	ROC-AUC	Mass MAE (GeV)	Mass r	p_T MAE (GeV)	p_T r
XCiT (finetune)	0.8840	0.9488	18.305	0.875	59.001	0.630
XCiT (scratch)	0.8805	0.9473	19.733	0.868	63.377	0.554
ViT (finetune)	0.8245	0.8968	22.595	0.807	67.018	0.449
ViT (scratch)	0.8620	0.9205	22.958	0.817	75.795	0.034

5.3 Analysis and Conclusions

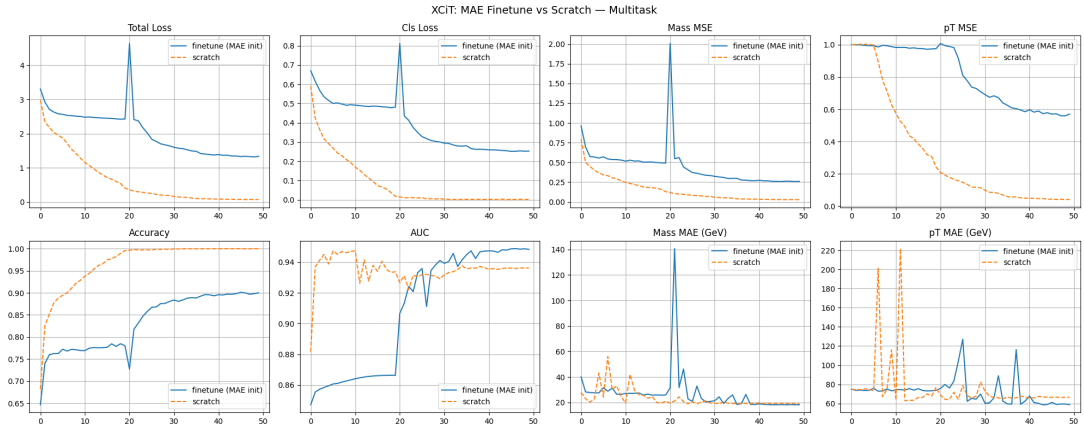

Figure 4: Training dynamics for finetune vs scratch. The discontinuity at epoch 20 corresponds to backbone unfreezing.

Classification

- Both models achieve **ROC-AUC** > 0.93.
- Finetune is slightly better than scratch: **0.9488 vs 0.9473**.
- The training curves show a clear **phase transition at epoch 20**.
- XCiT achieves performance comparable to or better than the standard ViT baseline under identical training settings.

Jet Mass Regression

- Finetune better: **MAE 18.30 vs 19.73 GeV**.
- Higher correlation: $r = 0.875$ **vs 0.868**.
- Indicates a strong **transfer from MAE pretraining**.
- The learned features capture **the geometry of the jet substructure**.

p_T Regression

- Finetune better: **MAE 59.00 vs 63.38 GeV**.
- Higher Correlation: $r = 0.630$ **vs 0.554**.
- Despite the weak signal strength, finetuning captures the residual energy-scale information.

Summary

- **MAE pretraining significantly improves regression**.
- Remains **competitive on classification**.
- Provides a better **accuracy vs interpretability trade-off**.

6 Proposed Timeline

Community Bonding Period

- Review linear attention literature and jet substructure observables with mentors.
- Review the relevant jet physics context (jet substructure, mass, and p_T) with mentors and document key assumptions and constraints that will guide model design and evaluation.
- Discuss the integration of linear attention methods into vision transformers.
- Finalize dataset structure and evaluation protocols.
- **Deliverable:** Agreed project plan and finalized experimental setup.

Week 1: Infrastructure and Baselines

- Finalize data pipelines for labeled and unlabeled datasets.
- Confirm dataset splits for training, validation and testing.
- Implement a standard ViT baseline.
- **Deliverable:** Working data pipeline and baseline ViT results.

Weeks 2–3: Linear Attention Architectures

- Implement XCiT from scratch in PyTorch.
- Benchmark classification against baseline architectures.
- Evaluate the models based on ROC-AUC, F1 score, training time, and parameter count.
- **Deliverable:** XCiT implementation with baseline comparison results.

Weeks 4–5: Multitask Pipeline

- Extend the backbone with parallel mass and p_T regression heads.
- Implement a two-step training protocol (frozen $\rightarrow$ end-to-end fine-tuning).
- Evaluate finetuned vs. scratch across all tasks.
- **Deliverable:** Multitask model with classification and regression results.

Week 6: Self-Supervised Pretraining

- Implement MAE pretraining with sparse-image adaptations (patch-normalized loss, activity weighting, reduced mask ratio).
- Evaluate downstream improvements from pretraining.
- **Deliverable:** Pretrained encoder and comparative finetuning results.

Midterm Evaluation. All core architectures are implemented and benchmarked. Initial multitask and pretraining results available for evaluation. Further improvements will be made according to feedback from the mentor(s).

Weeks 7–8: Interpretability

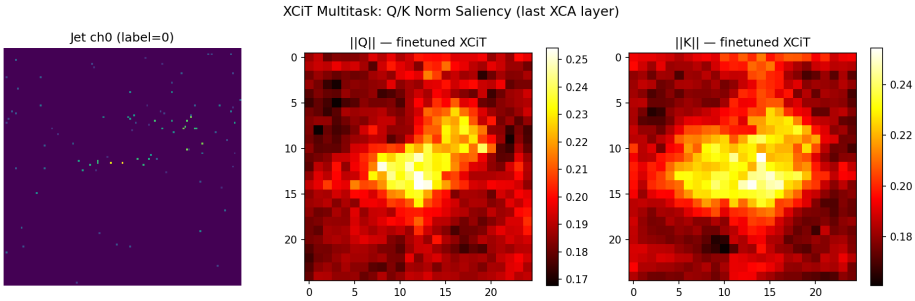

Figure 5: Q/K norm saliency maps from the final XCiT layer, showing attention concentrated on jet core regions.

- Visualize XCA Q/K norm saliency maps and LCM activations.
- Correlate latent features with physics observables (N-subjettiness, thrust, mass, p_T, etc.).
- **Deliverable:** Interpretability analysis linking learned features to physics observables.

Weeks 9–10: Documentation and Deliverables

- Refactor and document the codebase for clarity and reproducibility.
- Release model weights and training scripts.
- **Deliverable:** Public repository with reproducible pipeline and pretrained models.

Weeks 11–12: Buffer and Additional Experiments

- Conduct additional experiments and improvements.
- Address unforeseen technical challenges.
- Improve visualizations and analysis.
- Prepare the final report.
- **Deliverable:** Final report and complete experimental results.

7 Potential Challenges and Mitigation

Sparse-image MAE: Vanilla MSE reconstruction on sparse jet images trivially minimizes by predicting zeros. This is addressed by per-patch normalized targets, activity-weighted loss, and a reduced mask ratio of 60%, as validated in the preliminary results.

Linear attention instability: Denominator clamping in ReLU attention is dataset-dependent. Loss curves are monitored and $C_{\min}$ is tuned empirically; gradient clipping at 0.5 provides additional stability.

Multitask loss balancing: Regression targets are z-scored to bring losses to the same scale. Loss weights may be further tuned via grid search if needed.

p_T regression: Per-sample max-normalization does not make jet images fully p_T-invariant; residual energy-scale information survives through the image dynamic range. Future work will explore nonnormalized image variants or explicit total-energy channel injection to further improve p_T prediction.

Compute budget: Mixed-precision training (AMP), efficient XCA attention, and a moderate encoder depth (8 blocks) keep training feasible on a single GPU.

References

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